MAE 1001:
Bias in Dice: Your first experiment

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2025/

Prof. Kartik Bulusu, MAE Dept.

Graduate Teaching Assistant:
Preethi Siva Kumar (MAE Dept., MS student)

Learning Assistants:
Nathan Jannsen (Senior, MAE Dept.)
Olivia Falciani (Senior, MAE Dept.)
Shota Kakiuichi (Senior, MAE Dept.)

Tech support:
Rithik Gunuganti (CS Dept., MS student)

GW School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

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Photo: Kartik Bulusu

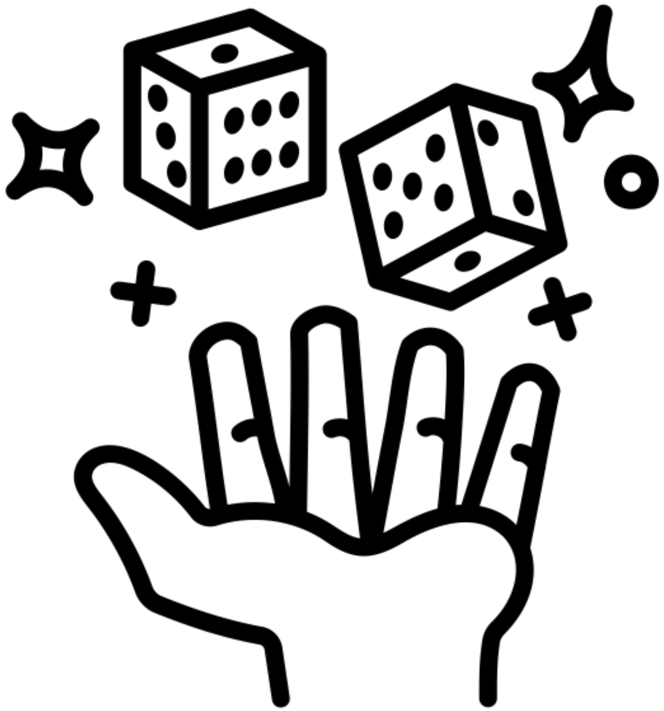
"Study hard what interests you the most in the most undisciplined, irreverent and original manner possible."

— Richard Feynman

What is Dice Bias?

Dice bias:

Happens when certain outcomes occur more often than chance would predict



Causes:

- Manufacturing flaws
- Uneven density
- Off-center markings
- Wear and tear

Source:
https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/center-of-gravity/
https://en.wikipedia.org/wiki/Center_of_pressure_(fluid_mechanics)
Shames, I., *Mechanics of Fluids*, McGraw-Hill, New York, U.S.A
https://www.grc.nasa.gov/www/k-12/WindTunnel/Activities/buoy_Archimedes.html
https://www.britannica.com/science/buoyancy
https://en.wikipedia.org/wiki/Buoyancy
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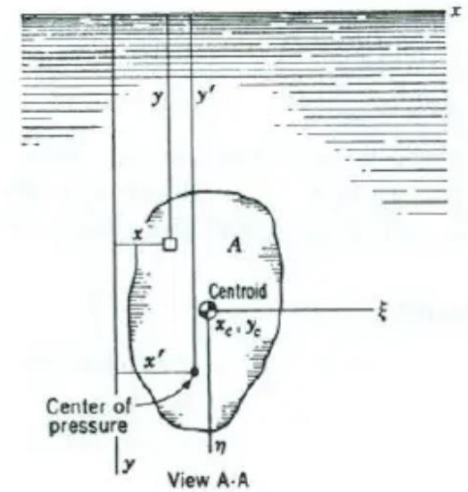
Preliminaries

Center of mass

The **center of mass** is a hypothetical point where the entire mass of an object may be assumed to be concentrated to visualize its motion.

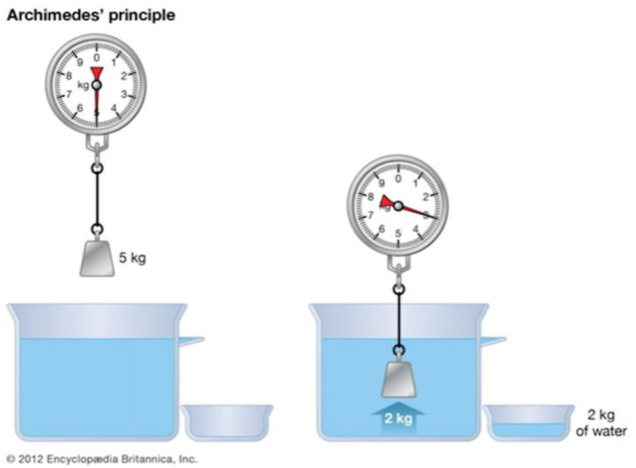
- Describes the motion of any object through space in terms of the **translation** and/or **rotation**
- **Application:** If the object is confined to rotate about some other point, like a hinge, we can still describe its motion.



Buoyancy principle:

An object floats when the upward buoyant force equals or exceeds the downward force from its weight.

The **buoyancy principle**—formulated as Archimedes' principle—states that the buoyant force acting on an object immersed in a fluid is equal to the weight of the fluid displaced by that object.



Interplay between Center of mass, Buoyancy and Dice

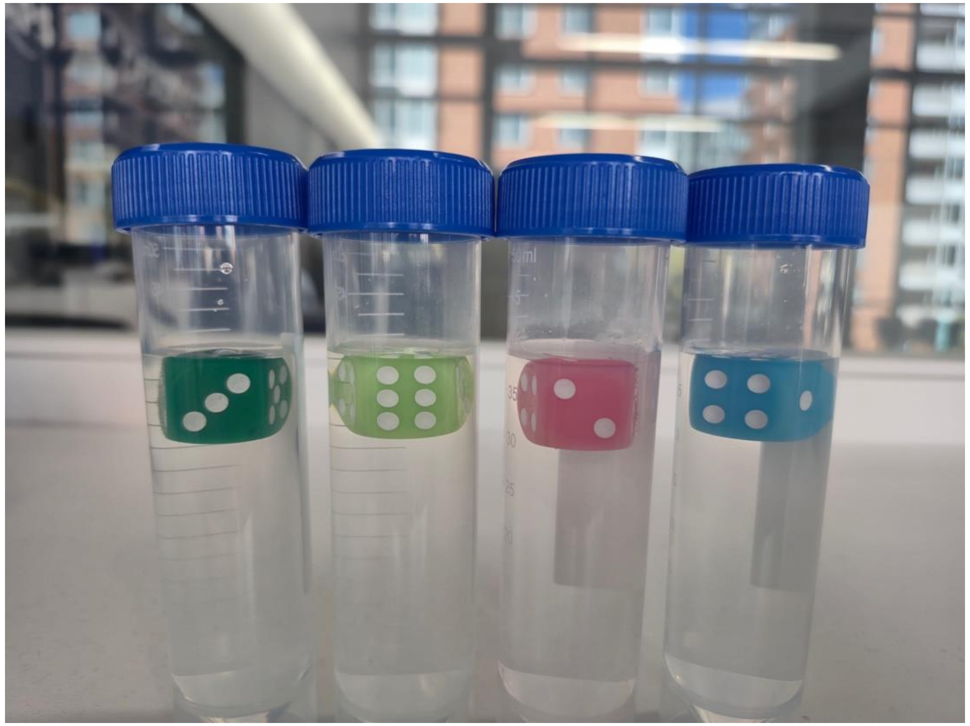
Physical principle:

- For a fair die, center of mass is at the geometrical center
- Non-uniform dice misbehave when floating due to uneven weight

Method:

- Float die in a saltwater/sugar solution to test center of mass location by observing which face points downward/upward.

A perfectly balanced die should show random orientation after several tests



Buoyancy of Float Test Method

Materials:

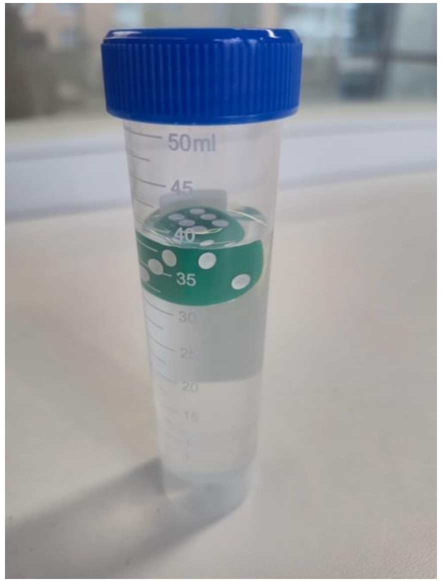
- Dice
- Water and salt (or sugar for higher density)
- Sealed centrifuge tube with a dice floating in it

Consistent results indicate off-center mass or bias in dice

Steps:

1. **Download the Python program** from the GitHub Repo on your computer: https://github.com/gwu-mae1001/fall2025_codes.git
2. **Launch Anaconda Navigator**
   1. **Locate the program**
   2. **Run the program** and **be ready to record your data** from Step 5
3. You will be given a centrifuge tube with a floating dice in it: **Shake the centrifuge tube.**
4. **Observe the dice as it floats up.**
5. **Record the face value** that shows up as the dice floats and rests on the fluid surface.
6. **Repeat the test several times for reliability.**
7. **Locate the data saved in a CSV file on your computer**

Discussion



- **How many of your felt that your dice was indeed biased?**
- **Did we do enough tests to confirm or bias or no-bias?**
- **What are the limitations of this experiment?**
- **How can we link the physical phenomena and the deeper analysis?**

- **Bouyancy reveals** a physical imbalance, but actual roll data is needed for quantification.
- **We need** a large dataset to link the physical phenomenon to mathematical analysis.
- **Assignment deliverable:** Upload your data file (CSV file) on BlackBoard to get a grade

We will repeat this test and apply statistical analysis using the chi-square test

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

Where O_i is observed frequency, E_i is expected (n faces)